

EMPLOYEE ATTRITION ANALYTICS

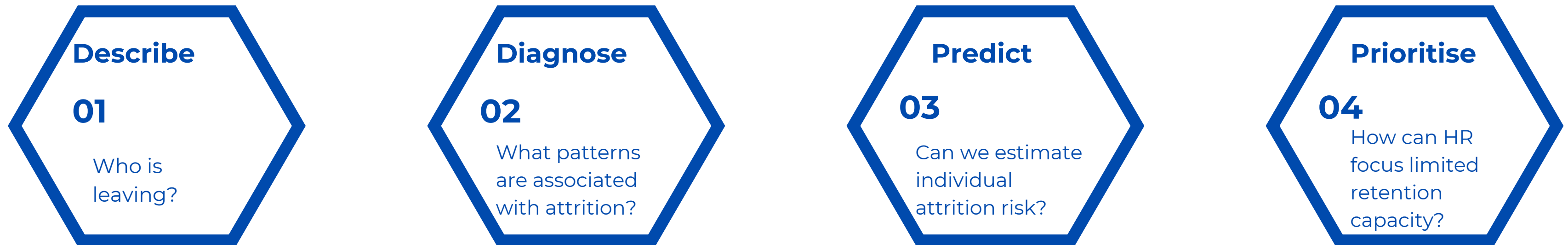
Understanding workforce turnover and identifying opportunities for proactive retention

People Analytics Case Study
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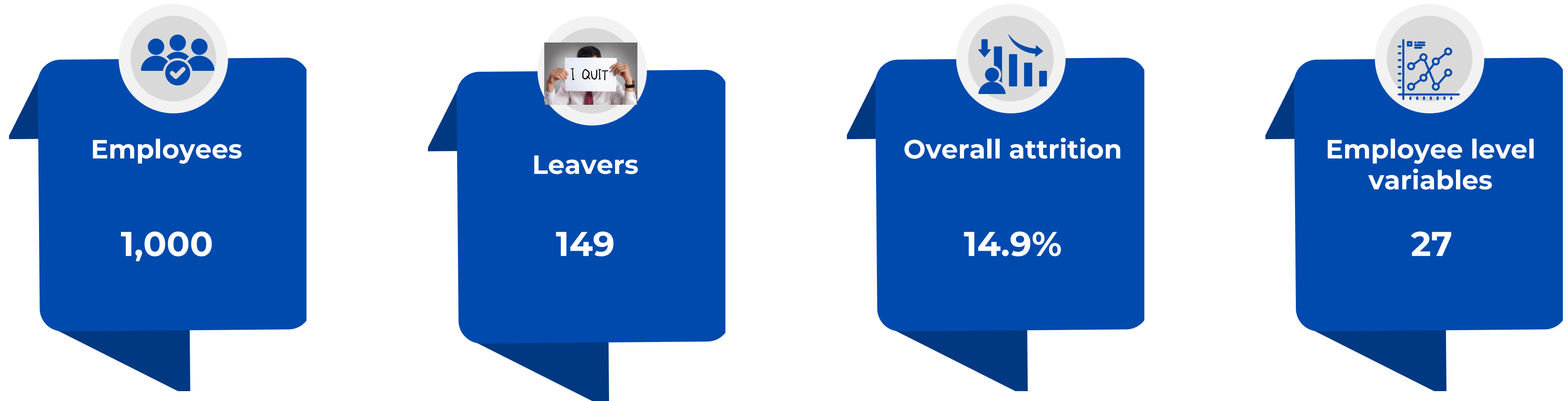
The Business Question

What is driving employee attrition, which workforce segments are most affected, and how can People Analytics support proactive retention?

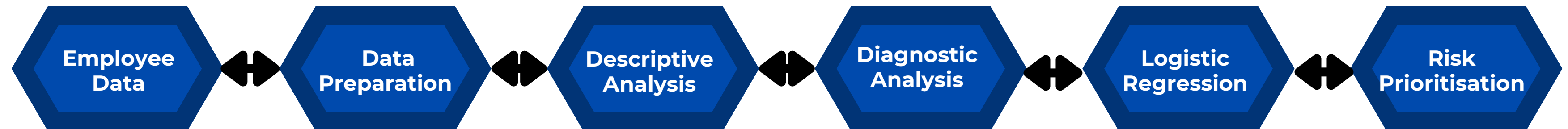


Data → insight → prediction → action

Data & Methodology



Analytical Process / Sequence

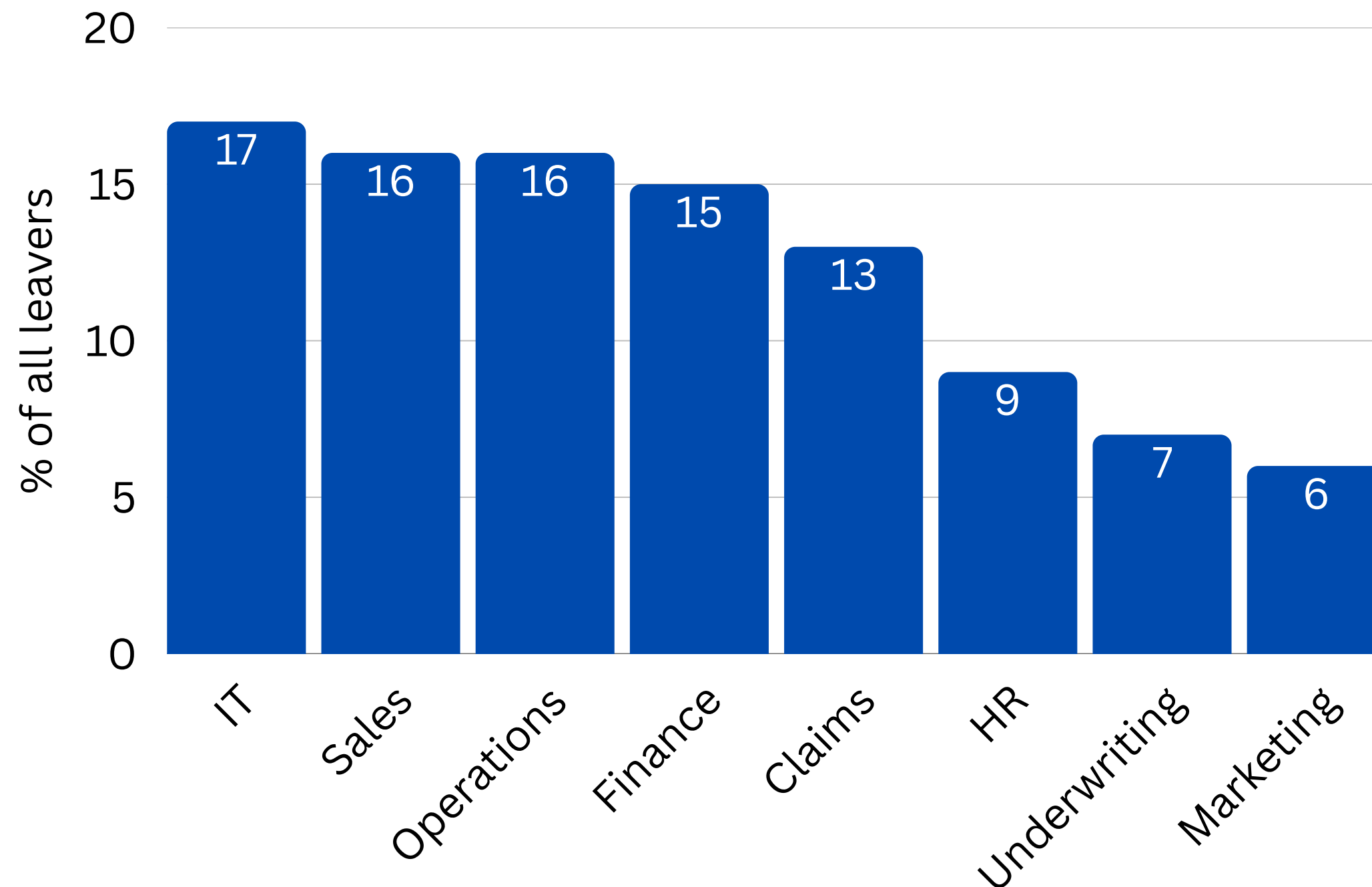


1. Workforce characteristics: gender, age, country, department, job level
2. Career & tenure: tenure, time in role, promotion, manager tenure
3. Reward: annual salary, compa-ratio, months since salary increase
4. Employee experience: engagement, absence, overtime, training
5. Work arrangement and performance

Descriptive Analysis — Who is leaving?

Leavers are concentrated across several large workforce segments.

Share of 149 leavers by department



Largest leaver share

IT 16.8%

of all leavers

Next highest

Sales 16.1%

Operations 16.1%

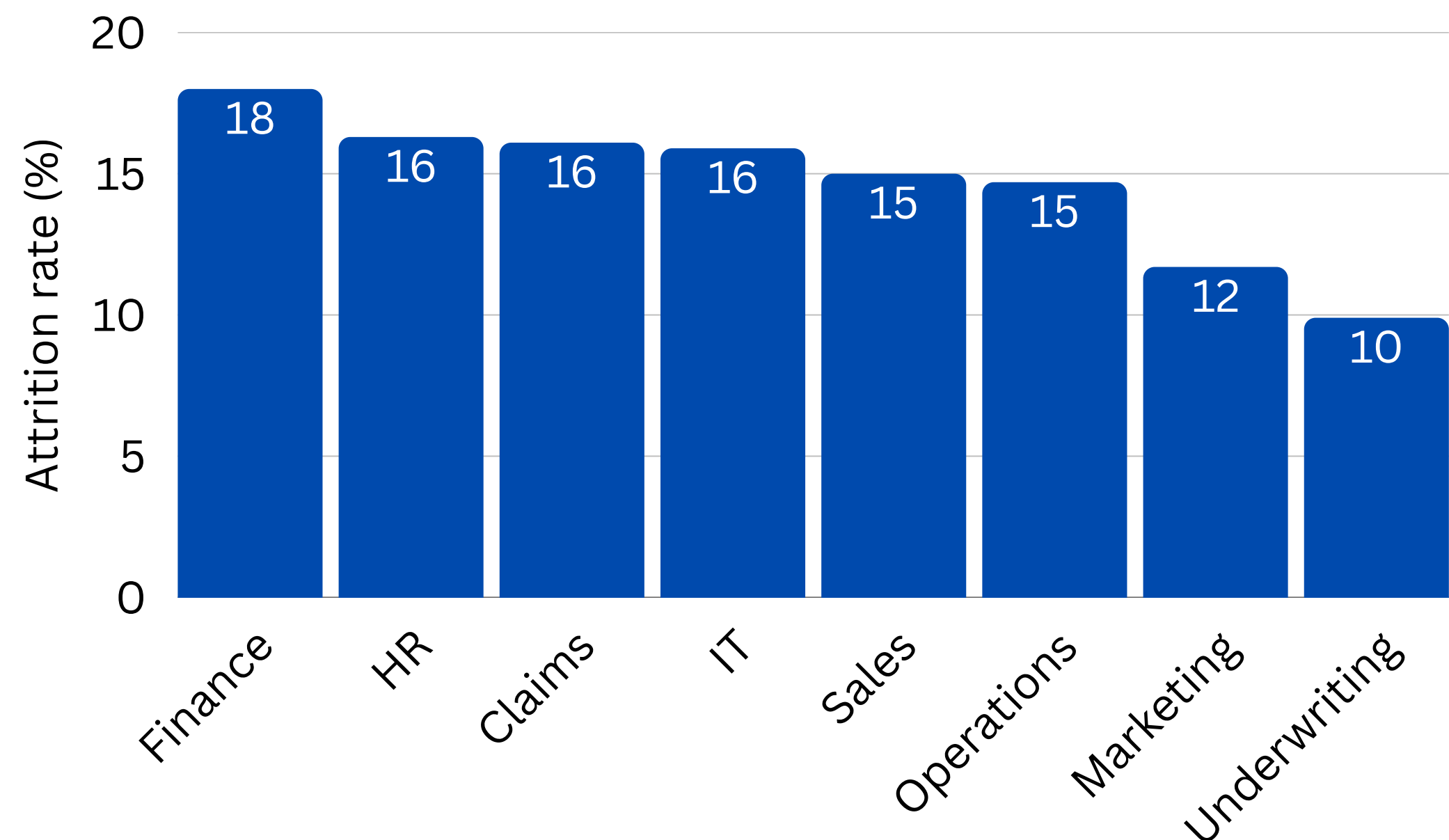
Leaver profile

- 60% of leavers were aged 32–45.
- 44% of leavers had 2–<5 years' tenure.
- 36% of leavers were Professional-level employees.
- 40% of leavers were based in the UK.

Attrition Analysis — Where Is Attrition Highest?

Attrition rate identifies segments that differ from the 14.9% workforce baseline

Department attrition rate



Higher-attrition populations identified

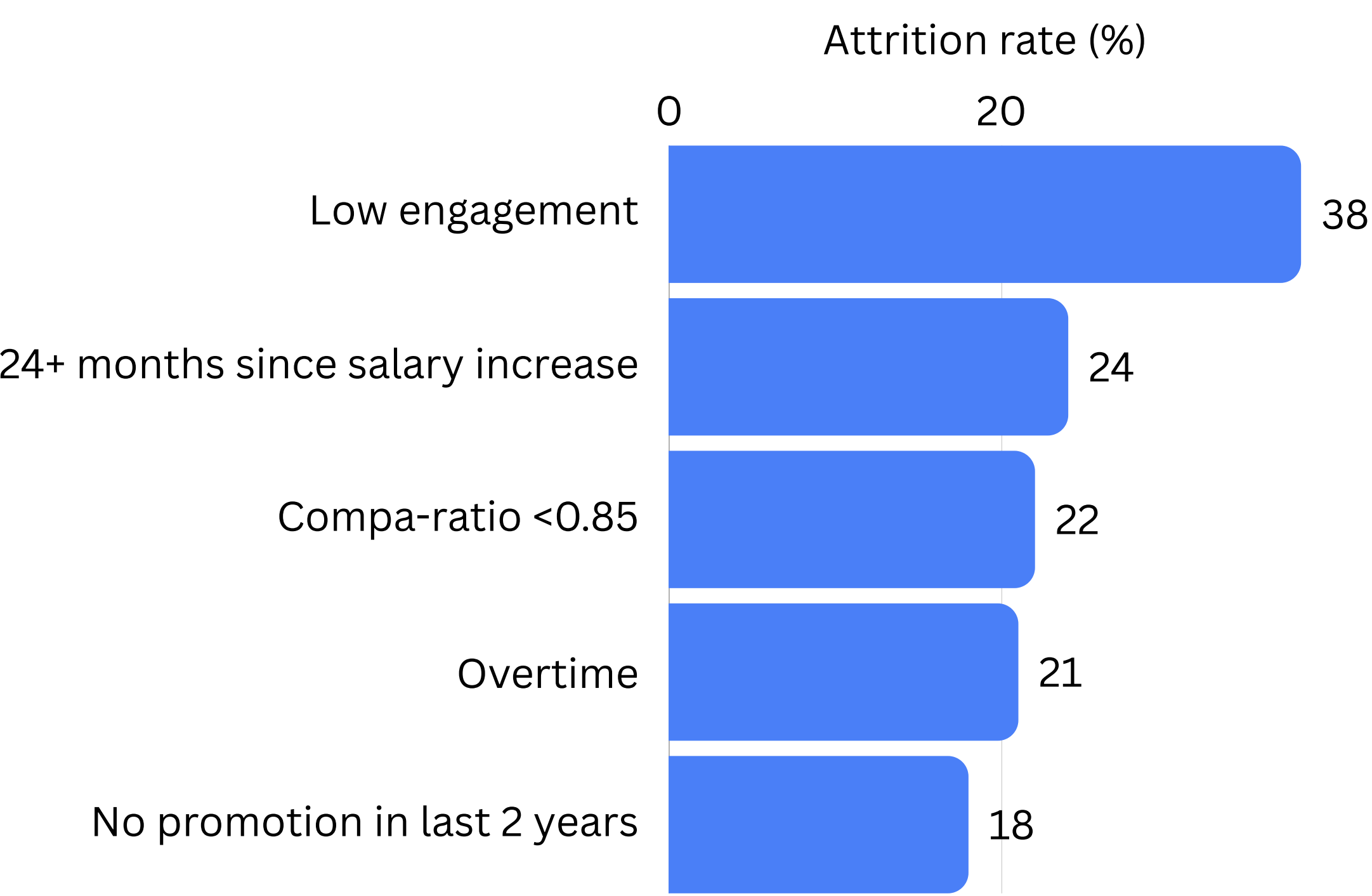
Canada	28.0%	+13.1pp vs overall
Entry level	21.0%	+6.1pp vs overall
Finance	18.0%	+3.1pp vs overall
Female	17.3%	+2.4pp vs overall
HR	16.3%	+1.4pp vs overall
Claims	16.1%	+1.2pp vs overall

The 60–73 group has a 50% rate but represents only 1% of leavers; small populations require caution.

Attrition rate measures the proportion of each workforce segment that left, rather than its share of all leavers.

Diagnostic Analysis — What is Associated With Attrition?

Cross-tab analysis highlights recurring patterns in engagement, reward, progression and workload



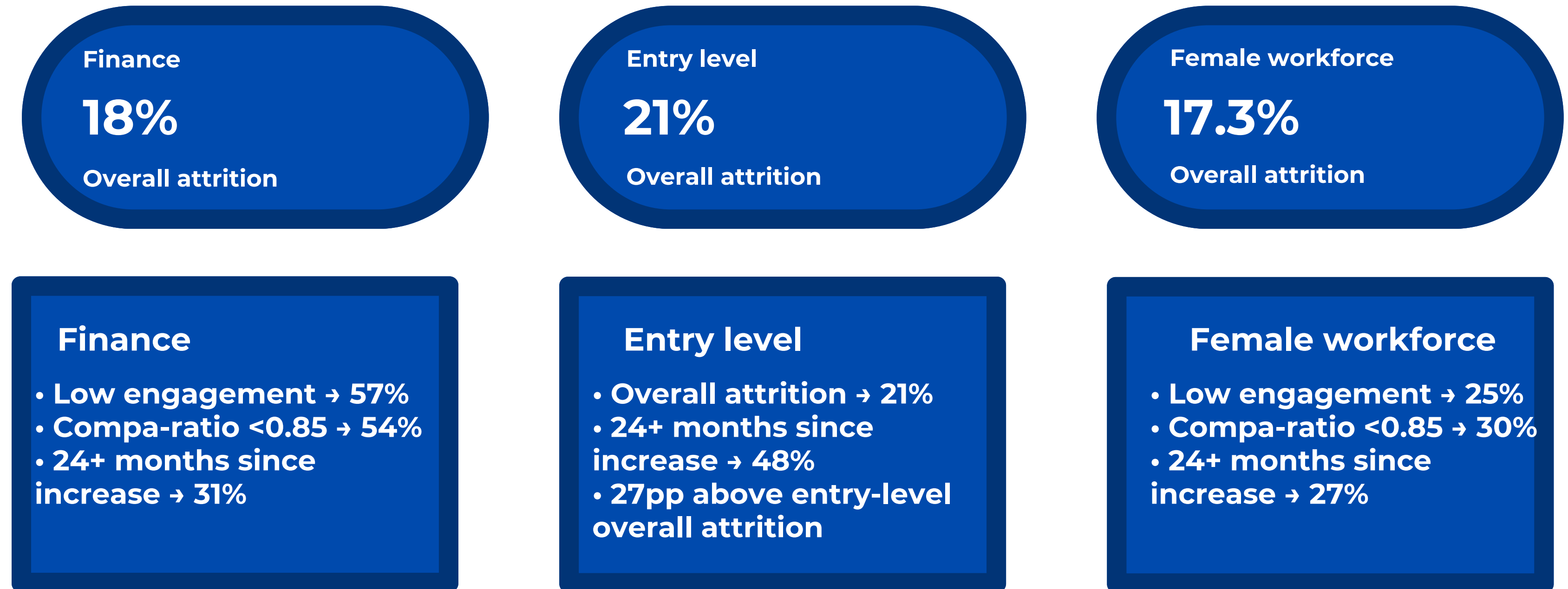
What stands out

- Low engagement: 38.0% attrition vs 9.6% for very-high engagement.
- 24+ months since salary increase: 24.3% vs 12.2% for 6–12 months.
- Compa-ratio <0.85: 22.0% vs 13.1% for 0.85–<1.00.
- Overtime: 20.7% vs 12.4% without overtime.
- No promotion: 17.7% vs 7.8% with promotion in the last 2 years.

These are observed associations in the dataset and do not establish causation.

Intersection Analysis — where do patterns become more concentrated?

Combining workforce segments with diagnostic factors creates a more actionable view



Use intersections to define questions for deeper investigation — not to label a group as inherently high risk.

Predicting Employee Attrition Risk

Logistic regression converts employee characteristics into a probability of attrition



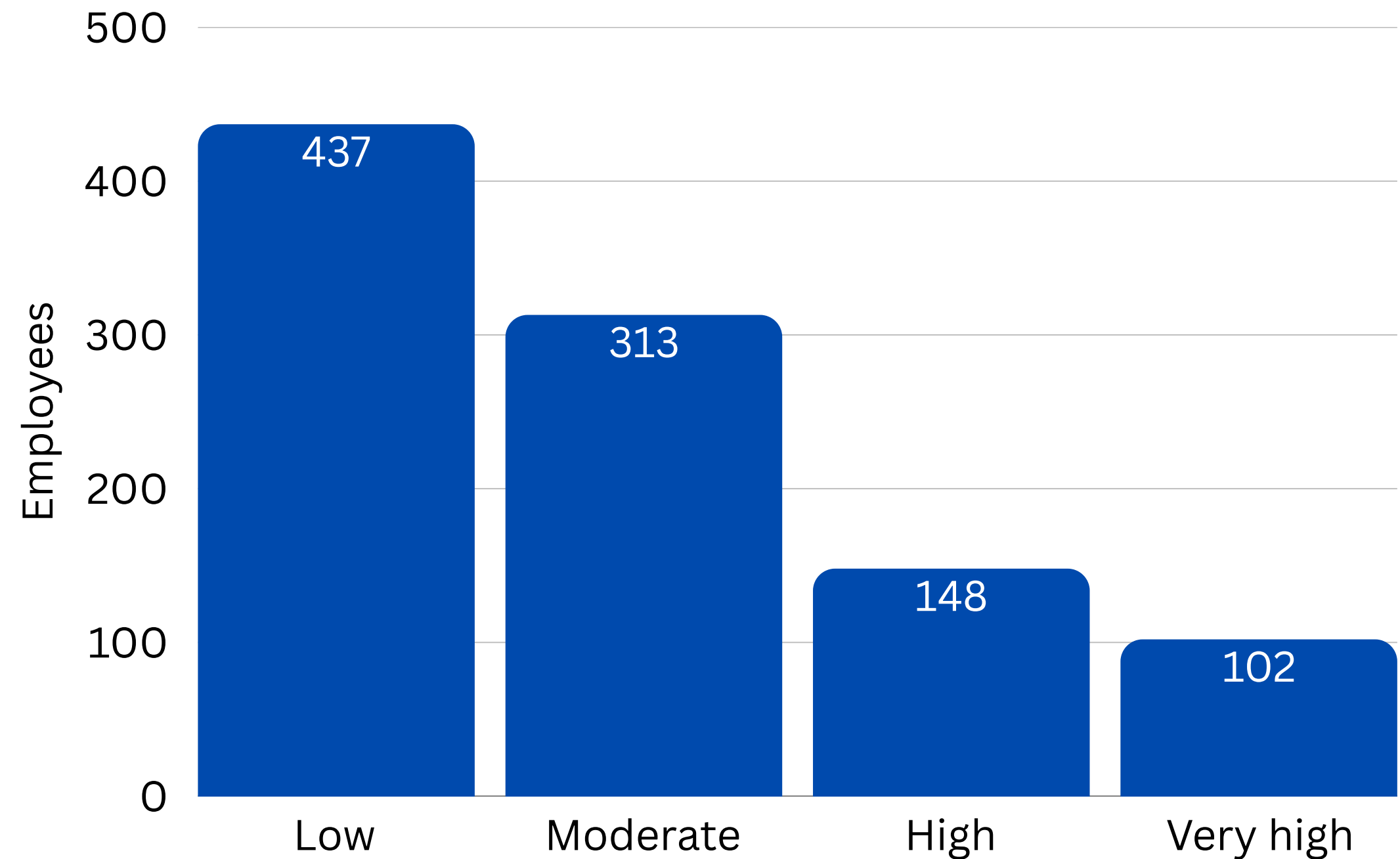
Model workflow



Predicted risk distribution

Working probability bands translate model output into an HR-friendly risk view

Employees by predicted risk band



Band validation

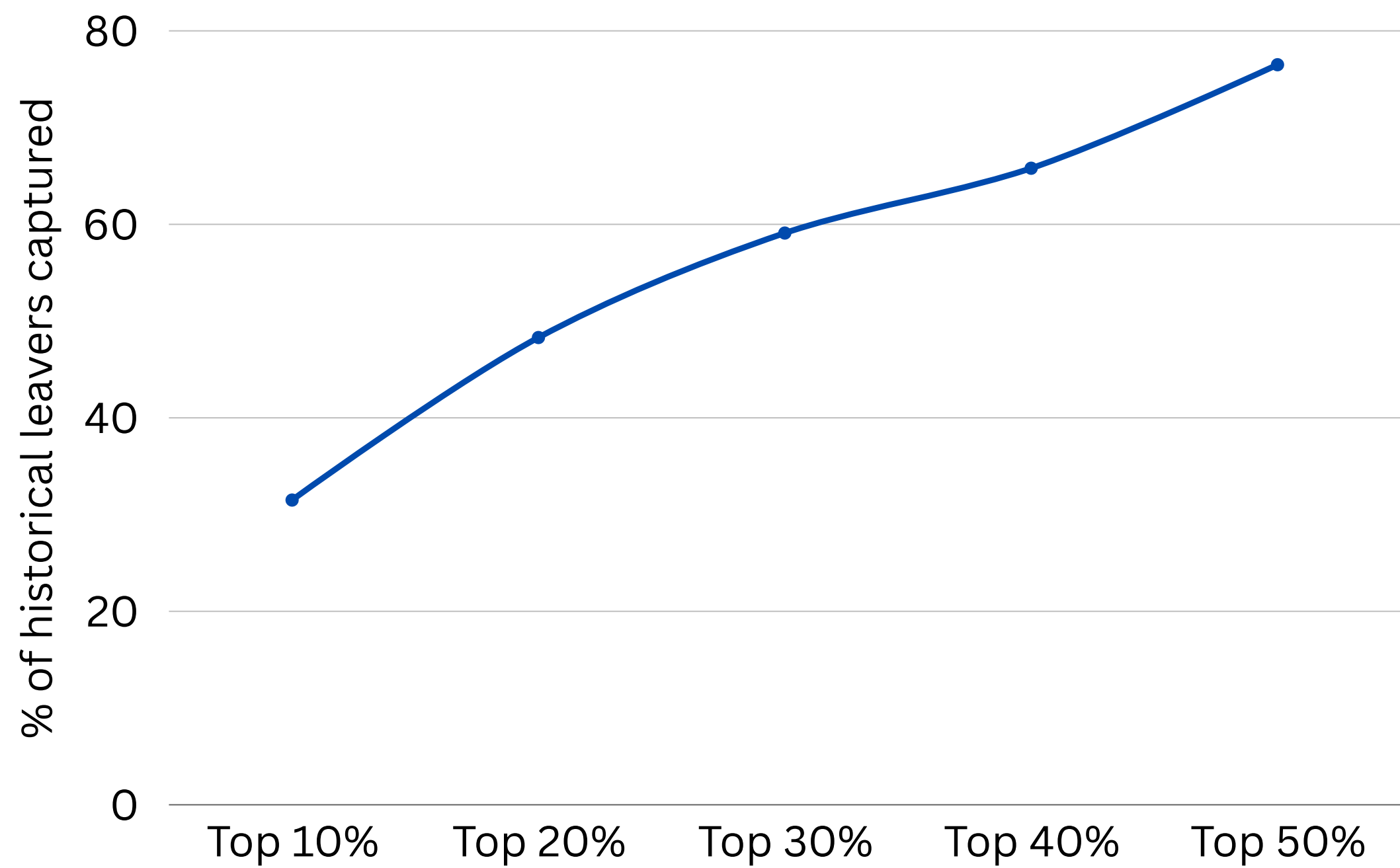
Low	437 employees	6.9% attrition	0.46× lift
Moderate	313 employees	12.1% attrition	0.81× lift
High	148 employees	22.3% attrition	1.50× lift
Very High	102 employees	47.1% attrition	3.16× lift

Observed attrition increases across the risk bands, supporting the use of the probability ranking for prioritisation. Thresholds are working bands and should be validated against future data.

Retention Prioritisation — How much of the historical leaver population is contained within the highest-risk groups?

Employees are ranked from highest to lowest predicted risk

Cumulative leavers captured



Prioritisation view

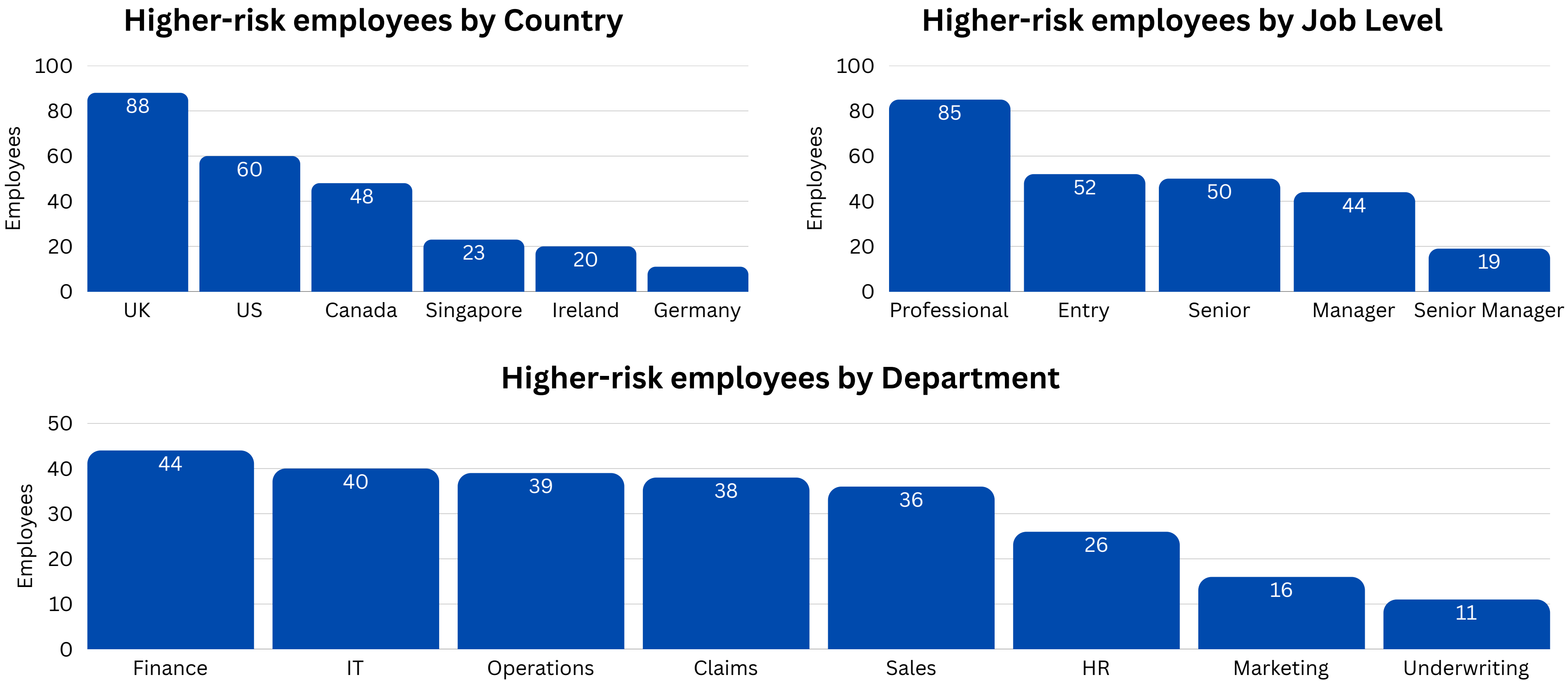
Top 10%	31.5% captured	3.15× capture-rate lift
Top 20%	48.3% captured	2.42× capture-rate lift
Top 30%	59.1% captured	1.97× capture-rate lift
Top 40%	65.8% captured	1.65× capture-rate lift
Top 50%	76.5% captured	1.53× capture-rate lift

Example: the top 10% of employees by predicted risk contains 31.5% of historical leavers.

This can help HR prioritise limited retention capacity; it does not mean every employee in the group will leave.

Higher-Risk Workforce Profile - (250 higher- risk)

Who Makes Up the Higher Predicted-Risk Population?



From Insight to Action

Recommendations are framed as questions and interventions to validate — not causal conclusions

Engagement

Low engagement is associated with 38% attrition.

Run targeted engagement analysis / focus groups to understand the underlying experience.

Reward

Compa-ratio <0.85 is associated with 22% attrition.

Review pay positioning and market competitiveness for affected populations.

Progression

24+ months since salary increase is associated with 24% attrition.

Review salary progression, career pathways and pay-review history.

Career

Entry-level attrition is 21%; 48% among those with 24+ months since increase.

Review early-career development, progression and reward pathways.

Workload

Overtime is associated with 20.7% attrition.

Investigate workload, resourcing and team-level overtime patterns.

Success measure: track intervention uptake, employee experience and subsequent attrition over time

Limitations & Next Steps

Model validation

The current model was developed using the full dataset.

Future development should use holdout data or cross-validation.

Causality

Observed associations do not establish that a factor causes attrition.

Risk thresholds

Risk-band thresholds should be validated against future outcomes and business capacity.

Small populations

Small workforce segments require cautious interpretation.

Responsible use

Individual risk scores should be treated as decision-support signals, not predictions that an employee will leave.

Model monitoring

Reassess model performance and calibration as workforce conditions change.

Key Takeaways



Key Takeaway 1

Attrition varies meaningfully across workforce segments.

Key Takeaway 2

Engagement, reward positioning, progression and overtime show notable associations with attrition.

Key Takeaway 3

The model can support risk-based prioritisation, subject to validation and responsible use.

THANK YOU

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